

Name: Nqubeko Zincume

Student Number: zncnqu001

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(putting all together)

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Lecturer: Prof Marcello Vichi

The Analysis of Chlorophyll-a Distribution and Seasonal Variation in the Benguela Region, Southwest cost of South Africa

Introduction

The Benguela Region is one of the major Eastern Boundary current, located off the southwest coasts of Southern Africa and is bounded by warm water at both its equatorward and poleward margins (Ragosha *et al.*, 2022), extends along the west coast of South Africa near Cape Town and the Cape Peninsula. The region is strongly influenced by the Benguela Current and its associated upwelling processes, making it an ideal location to study spatial and seasonal patterns in phytoplankton biomass.

According to Ragosha et al. (2022), the Benguela upwelling system plays a crucial role in moderating the climate of southwestern Africa, providing a rich source of nutrients that supports a highly productive marine ecosystem. Coastal upwelling in this region is driven by persistent southerly winds that transport surface waters offshore through Ekman transport, allowing cold, nutrient-rich waters from deeper layers to rise to the surface (Shannon & Nelson, 1996). Additionally, bathymetry shapes ocean circulation and the intensity of upwelling, as the continental shelf and shelf break influence the distribution of nutrient inputs and, consequently, phytoplankton productivity within the Benguela ecosystem (Martin *et al.*, 2024). Therefore, these nutrients stimulate phytoplankton growth, which can be observed through high concentrations of chlorophyll (Wasmund et al., 2024) derived from satellite ocean colour data.

Aim

This study aims is to visualise and analyse satellite-derived ocean colour data from the ESA Climate Change Initiative (ESA-CCI) Ocean Colour climatology, using python by examining the seasonal variability of chlorophyll-a concentrations in a selected coastal region of the Benguela Current system.

Objectives

1. Map the bathymetry of the selected Benguela Current region to visualise the seafloor topography and identify important features such as the continental shelf and slope.
2. Calculate and visualise the mean annual distribution of chlorophyll-a to examine the spatial patterns of phytoplankton biomass in the region.
3. Analyse the monthly climatology of chlorophyll-a concentrations to investigate seasonal variability in phytoplankton productivity.
4. Compare the seasonal cycle of the regional mean chlorophyll concentration with the chlorophyll time-series from a selected grid point located near an area of high productivity.

Methods

Study Area

The study area chosen for this exercise approximately (16°E–20°E and 31°S–35°S) lies along the west coast of South Africa near Cape Town and the Cape Peninsula.

Datasets

This study utilized a secondary dataset. Two main datasets were used in this study: satellite-derived chlorophyll data from the ESA Climate Change Initiative (ESA-CCI) Ocean Colour project and bathymetry data obtained from the Global Multi-Resolution Topography (GMRT). The period of the data was from (1997/09 to 1997/12 and 1998/01 to 1998/08).

Data procedure

A spatial resolution of approximately 4 km was set using the GMRT data to determine the bathymetry dataset was used to visualise the seafloor topography of the selected region and identify important features such as the continental shelf and shelf break. The data were derived from satellite ocean colour observations and processed as part of the ESA Ocean Colour Climate Change Initiative (OC-CCI). The dataset contains monthly climatological means, allowing the analysis of both spatial patterns and seasonal variability in phytoplankton productivity.

Data visualization analysis

All data processing, visualization, and analysis were performed using Python, by importing libraries such as xarray for managing multidimensional datasets, matplotlib and Cartopy using the PlateCarree Projection for geospatial plotting, and numpy for numerical operations. It was used to calculate the annual mean, seasonal mean, monthly mean, and the differences between the region cycle and point cycle mean, which then produced all the maps' plots and the time-series graph.

Results

The Benguela Region Bathymetry

The results indicate that the bathymetry map of the selected Benguela Current region illustrates the seafloor topography along the west coast of South Africa. It clearly maps the continental shelf and shows the transition to deeper ocean waters along the continental slope. In this area, the continental shelf is relatively narrow, with the shelf break occurring at approximately 200 m depth. Therefore, these bathymetric features play a crucial role in shaping coastal upwelling processes that occur along the continental shelf and shelf break, bringing nutrient-rich waters to the surface.

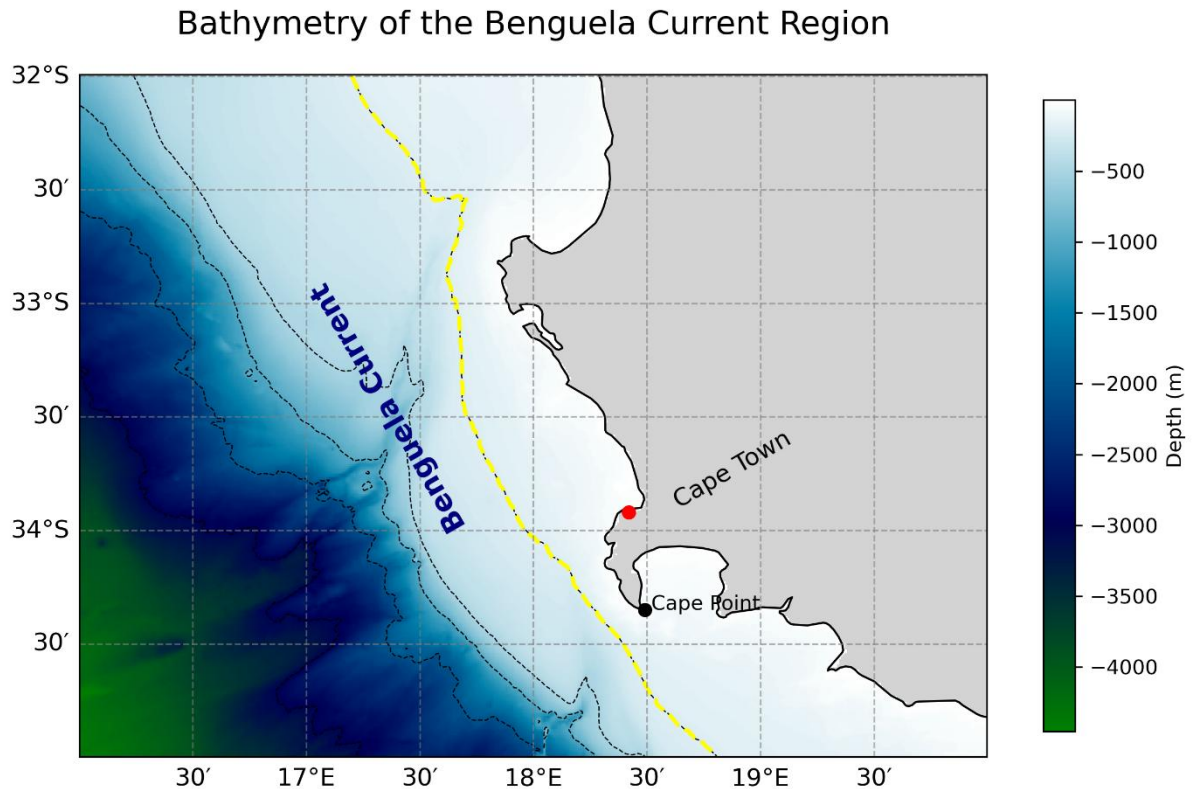


Figure 1: Bathymetry of the Benguela Current Region (16°E–20°E, 31°S–35°S), along the west coast of South Africa, with two geographical location, Cape Town and Cape Point. The map illustrates the seafloor depth from the GMRT dataset, using colour map 'ocean'. The lighter colour (white light blue) indicates the shallow water on the continental shelf is represented by dashed contours(yellow)- the continental shelf break approximately (200m) depth, while the darker blue colour (1000-3500m), indicates the deep offshore water, further changing to green colour showing the deeper part of the ocean (3500-500m).

Chlorophyll annual mean

The mean annual chlorophyll map shows a clear spatial pattern in phytoplankton biomass across the region. Bathymetric context highlights the continental shelf and slope, which play a key role in shaping coastal upwelling processes. Higher chlorophyll concentrations are observed near the coast approximately (10.0 mg.m^{-3}), particularly along the continental shelf, while lower concentrations occur offshore in deeper waters between ($0.193 - 0.072 \text{ mg.m}^{-3}$). This pattern illustrates the effects of the Benguela upwelling system, which enhances nutrient availability and phytoplankton productivity in coastal areas.

Chlorophyll Annual Mean Benguela Current Region

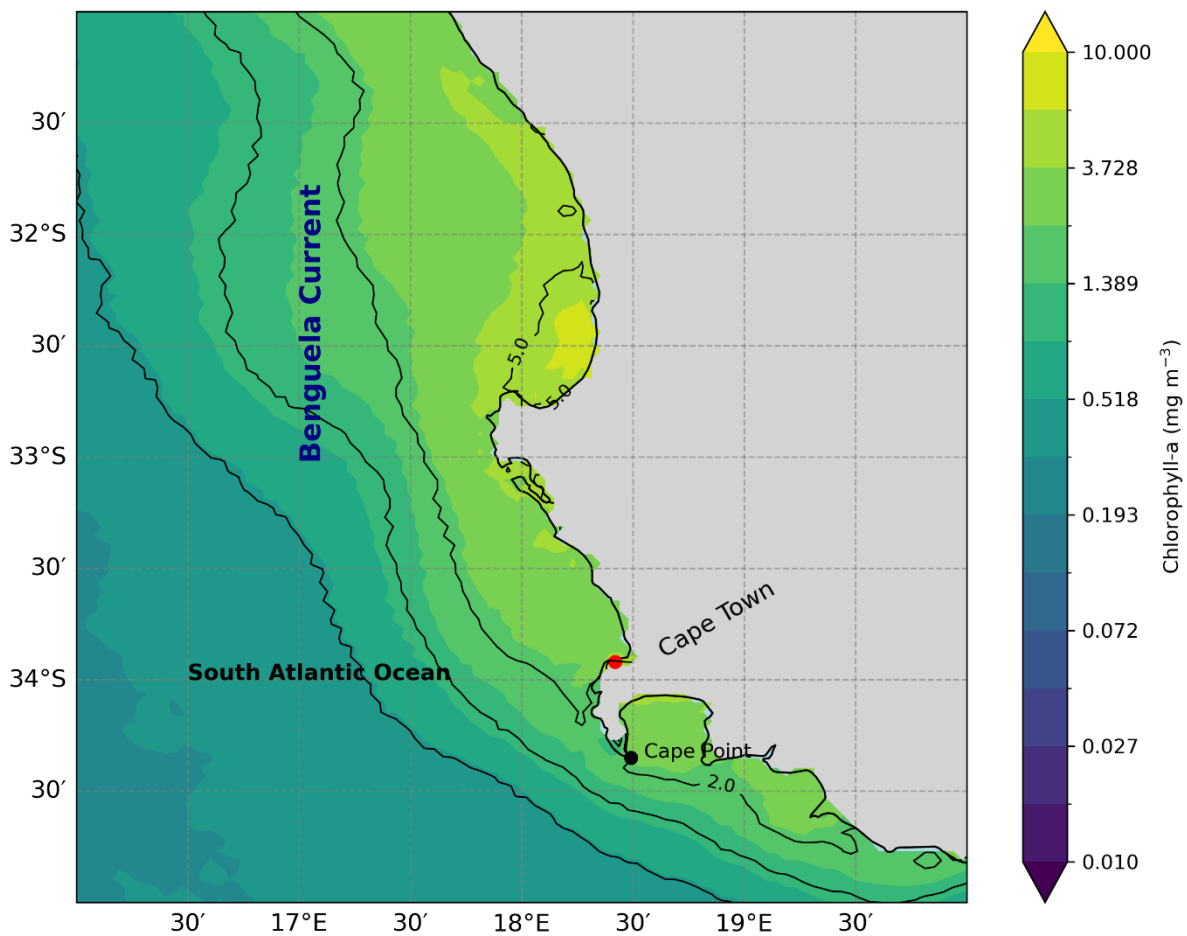


Figure 2: Annual mean chlorophyll-a concentration in the Benguela Current region (16°E–20°E, 31°S–35°S), with two geographical location, Cape Town and Cape Point. Derived from ESA-CCI Ocean Colour data, chlorophyll concentrations are shown on a logarithmic colour scale in units of mg m^{-3} . The map illustrates spatial variability in phytoplankton biomass along the west coast of South Africa, with elevated chlorophyll concentrations near the coast approximately (10.0 mg.m^{-3}), further moving toward the deep part of the ocean, chlorophyll concentration decreases between ($0.193 - 0.072 \text{ mg.m}^{-3}$).

Monthly Chlorophyll Climatology

The monthly chlorophyll climatology concentrations in this region shows significant seasonal variation of the phytoplankton biomass. Relatively high chlorophyll concentrations are particularly observed during the spring and summer months (September to February), especially near the coastal areas, where levels can exceed 10 mg m^{-3} in late spring. While offshore water moving towards the deep part of the ocean indicates lower chlorophyll concentration approximately between ($0.2 - 0.9 \text{ mg.m}^{-3}$) during the winter months (June–August). The patterns highlight the significance of upwelling intensity and seasonality in shaping phytoplankton biomass in the southwest.

Monthly Chlorophyll Climatology in the Benguela Current Region)

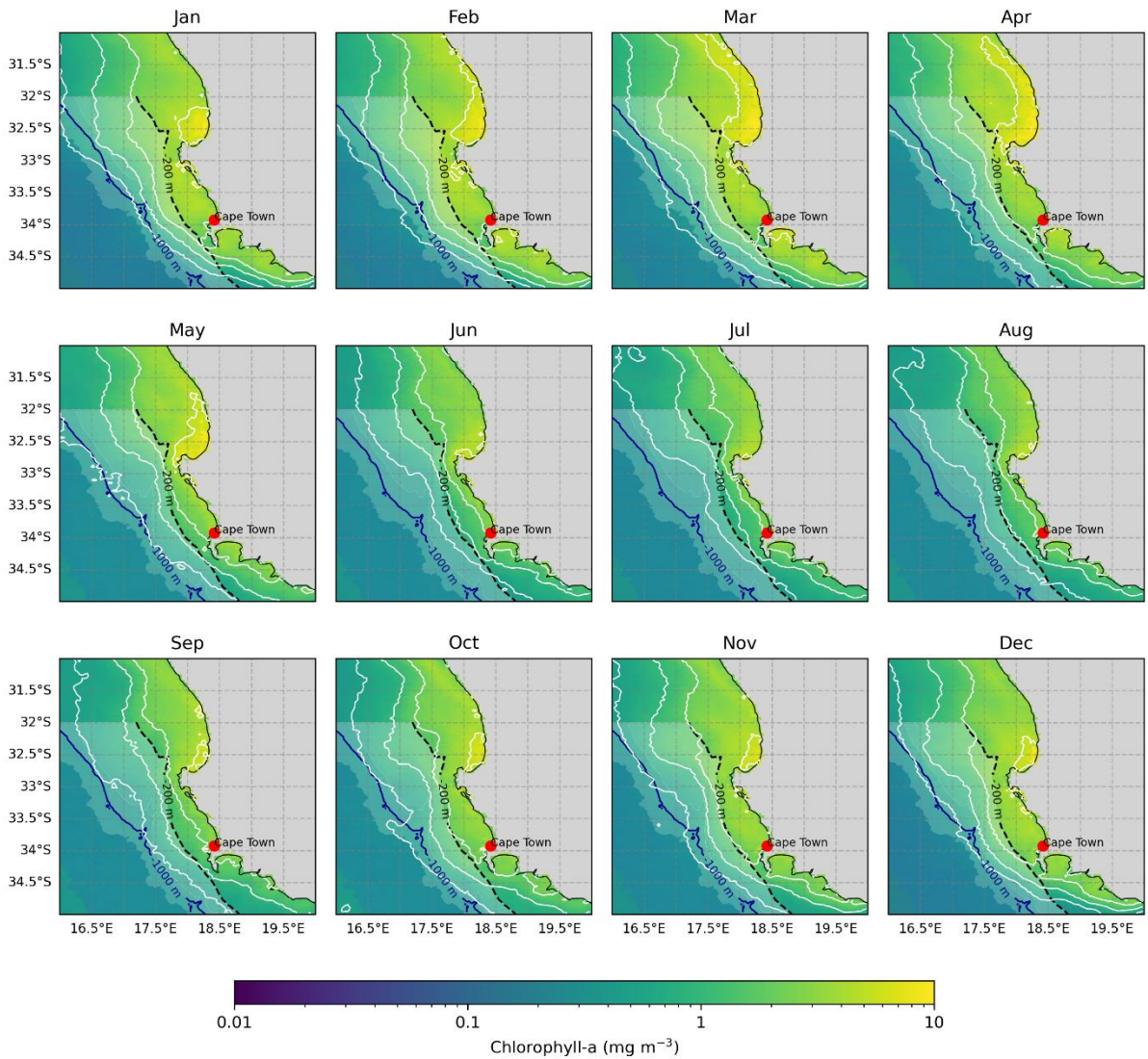


Figure 3: Monthly climatology of chlorophyll-a concentration (mg m^{-3}) in the Benguela Current region (31.5°S–34.5°S, 16.5°E–19.5°E). Derived from ESA-CCI Ocean Colour data, chlorophyll concentrations are shown on a logarithmic colour scale in units of mg m^{-3} . The continental shelf breaker is represented by dashed contours (black) approximately 200m in depth and the continental slope breaker represented by a solid line (dark blue) at 1000m depth. Maps show spatial distribution from January to December, Cape Town (red dot). The colour scale ranges from 0.01 mg m^{-3} (purple) to 10 mg m^{-3} (yellow). Elevated chlorophyll concentrations are observed the nearshore, towards the deep part of the ocean it was relatively lower.

Comparing regional mean and single point mean chlorophyll-a concentration

The mean seasonal cycle of chlorophyll-a concentration shows a clear and distinctive difference between the regional mean cycle and the single-point mean cycle. The regional mean remains relatively stable throughout the year, fluctuating only slightly between 0.5 and 0.8 mg m^{-3} , reflecting a relatively constant variation. While, the single point mean indicates pronounced seasonal fluctuations, at its maximum point in April (8.8 mg m^{-3}) and minimum in August (2.8 mg m^{-3}). Between January and April, chlorophyll concentration increased markedly from approximately (6 to 9.8 mg m^{-3}), followed by a gradual decline from April to August (8.8 to 2.8 mg m^{-3}). Concentrations then increase again toward the end of the year, reaching about 5.5 mg m^{-3} in December.

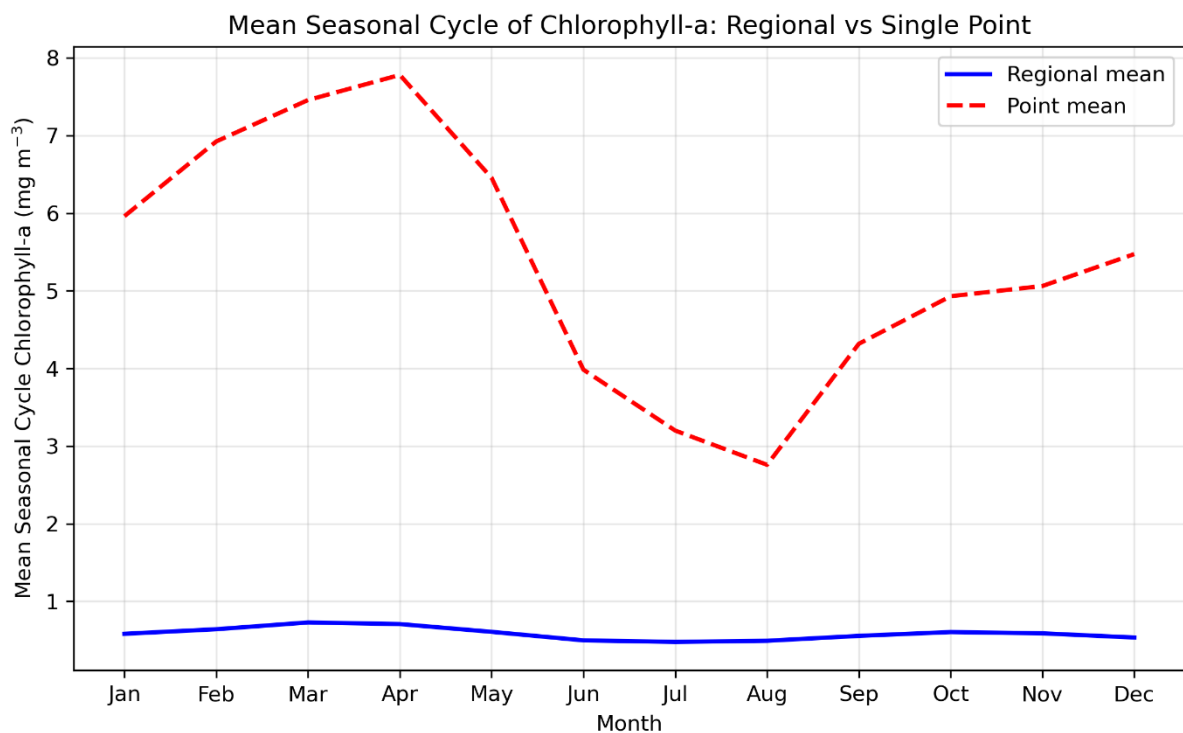


Figure 4: Mean seasonal cycle of chlorophyll-a concentration (mg m^{-3}) comparing a regional average (solid blue line) with a single-point mean (dashed red line) in the Benguela Current region. The regional mean remains relatively stable across months (0.5 and 0.8 mg m^{-3}), while the single point mean shows pronounced seasonal variability, maximum in April (8.8 mg m^{-3}) and declining to a minimum in August (2.8 mg m^{-3}).

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